

SO project - Stochastic Power Generation Planning

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February 25, 2025

Problem statement

In this project we consider the task of selecting power generators that minimize the total cost of supplying enough electricity to satisfy regional demand, which is [random](#).

To begin, suppose that we wish to select the optimal sizes of $J = 4$ different types of generators: 1 gas fired, 1 coal fired, 1 nuclear, and 1 oil. The annualized capital cost ($\text{€}/MWh$) for the acquisition of a generator of type $j = 1, \dots, J$ is given by c_j^{inv} , while the cost of producing a unit of energy ($\text{€}/MWh$) is given by c_j^{prod} . Note that while decisions regarding electricity generation may be postponed until regional demand is known, decisions regarding the acquisition of generation capacity (construction of the power plants) cannot be postponed. Thus, we see a natural [two-stage](#) progression of the decisions that are taken.

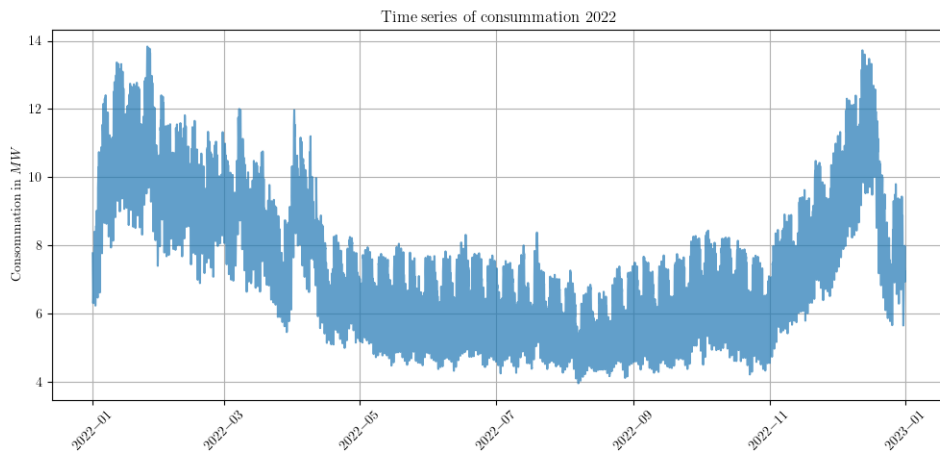


Figure 1: Chronological load curve of 2022

The regional demand for electricity is usually cyclical and the chronological load may be represented as shown in Figure 1. For the purposes of capacity planning, the chronology is

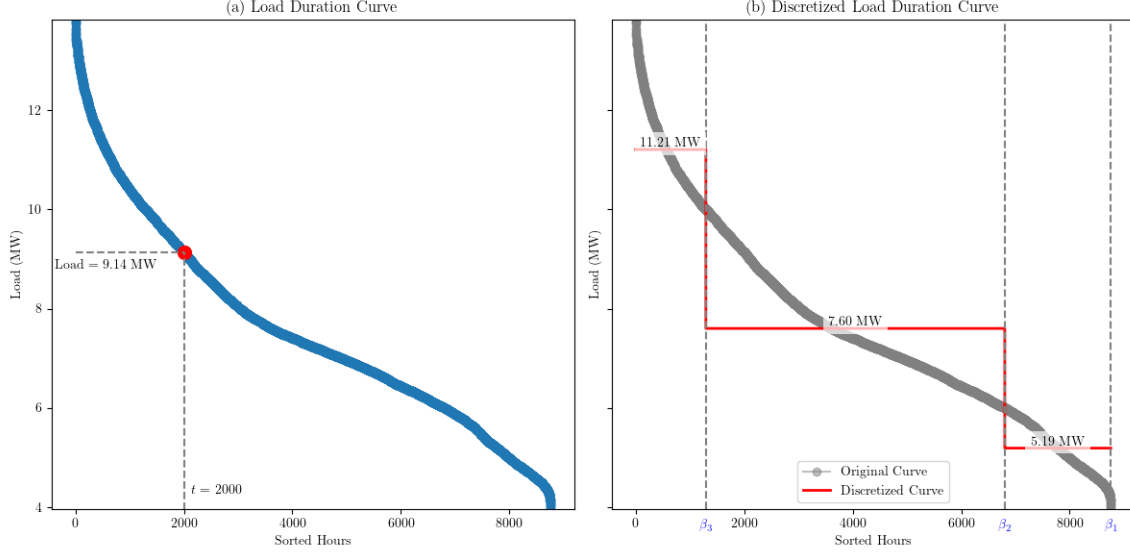


Figure 2: Load Duration Curve - LDC.

not as important as the duration of the load. A load duration curve (LDC, see Figure 2(a)) is a reordering of the chronological load curve, describing the number of hours in the year that load was greater than or equal to a given level. For a given load level (e.g., Load= 9.14MW), the horizontal axis provides the number of hours (2000 hours) during the year that the load exceeds this load. The area under the LDC gives the total energy consumption during the year.

For planning purposes, the LDC is often discretized into several steps as an approximation, as shown in Figure 2(b), with each segment representing the different load patterns:

- Base load: 0 – 5.19MW, lasts for 8760 hours (whole year)
- Medium load: 5.19 – 7.6MW, lasts for 6803 hours
- Peak load: 7.6 – 11.21MW, lasts for 1292 hours

Here we consider I steps of the discretized LDC. For $i = 1, \dots, I$, we require the step sizes (height of load block), D_i in MW, and the load duration (width of load block) β_i in hours. We assume that the duration of the steps, β_i , $i = 1, \dots, I$, which depend on characteristics such as the number of workdays vs. weekend days, summer vacations, etc. during the planning period are predictable. However, the step sizes, which correspond to the actual demand for power during the i^{th} load segment will vary in response to unpredictable factors in the long term (e.g., climate change, consumption behavior shift, etc.). Thus, these demands D_i for electrical power are treated as [random variables](#), whose probability distributions can be inferred from historical data (provided in Table 1). The decision variables are identified as

- x_j = MW of generation capacity type $j = 1, \dots, J$.

- $y_{ij} = MW$ of demand segment $i = 1, \dots, I$ served using generator type $j = 1, \dots, J$.
- $p_i = MW$ of load shedding in block $i = 1, \dots, I$.

A minimum capacity of P is imposed:

$$\sum_{j=1}^J x_j \geq P$$

Notice that the total power that can be obtained from a generator with capacity x_j is

$$\sum_{i=1}^I y_{ij} \leq x_j, \quad \forall j = 1, \dots, J$$

Moreover, the load duration curve demand must be satisfied for each block (load shedding p_i must be introduced if supply y_{ij} is insufficient):

$$\sum_{j=1}^J y_{ij} = D_i - p_i, \quad \forall i = 1, \dots, I$$

The total cost of acquiring capacity and producing electricity is for a fixed LDC is:

$$\sum_{j=1}^J c_j^{inv} x_j + \sum_{j=1}^J \sum_{i=1}^I \beta_i c_j^{prod} y_{ij} + \sum_{i=1}^I \beta_i c_i^{penl} p_i$$

Your tasks

- 1) *Two-stage SP model formulation:* Formulate the power generation planning problem as a two-stage stochastic recourse model.
- 2) *Model implementation and solution:* With the given distributions, solve your model (SAA, SAA with the Benders L-Shaped method, stochastic subgradient methods, etc). Report the purchased generation capacity and the total expected cost (investment + production + shedding penalty). You need to determine the number of samples N drawn from the distributions by yourself. (Optional: Instead of modeling this problem with load shedding penalty, you can also model this problem with chance constraint, e.g., the probability of demand satisfaction should be at least $1 - \gamma$ over the distribution of D . This can also help you determine the number of samples N .)
- 3) *Risk-averse formulation:* Formulate the problem as a two-stage stochastic linear programming model with risk-averse measure $\mathbf{CVaR}_\alpha$, instead of using the expected generation cost, in the objective function.

- 4) *Risk-averse model implementation and solution:* With the given distributions, solve your risk-averse model with different confidence levels α employing the decomposition algorithm you developed in task 2). You need to determine the number of samples N drawn from the distributions by yourself.

Data

- $J = 4$ different types of power generators, representing coal, gas, nuclear, and oil for $j = 1, 2, 3, 4$, respectively
- $I = 3$ steps of the discretized LDC
- $P = 15$ is the minimum capacity to be installed
- $c^{inv} = (16, 5, 32, 2)^\top$ in €/MWh are the investment costs for capacity installation
- $c^{prod} = (25, 80, 6.5, 160)^\top$ in €/MWh are the production costs
- $c^{penl} = (180, 200, 220)^\top$ in €/MWh are the load shedding costs
- $\beta = (1, 0.8, 0.25)^\top$ are the load curve weights (Here we can use normalized values of β in hours)
- The demand distributions, assumed independent, are as Table 1.

Load block	Fitted probability distributions
D_1	Lognormal($1.7, 0.1^2$) truncated in $[4, 6]$
D_2	Lognormal($1, 0.2^2$)
D_3	Lognormal($0.2, 0.6^2$)

Table 1: The demand distributions

Submission

You are provided with a [submission.ipynb](#) notebook file that contains necessary instructions. We recommend the [modeling language](#) cvxpy for its tractability for parametric optimization and the [solver](#) gurobi for its efficiency and academic license. However you can choose the modeling tool and the solver of your interest. Finish the task indicated in each block of the notebook and run it. Keep the running log of the notebook. Send the notebook to the email address xie.zhuojun@centralesupelec.fr and name the subject as [SO-2025-P1-Your Name].